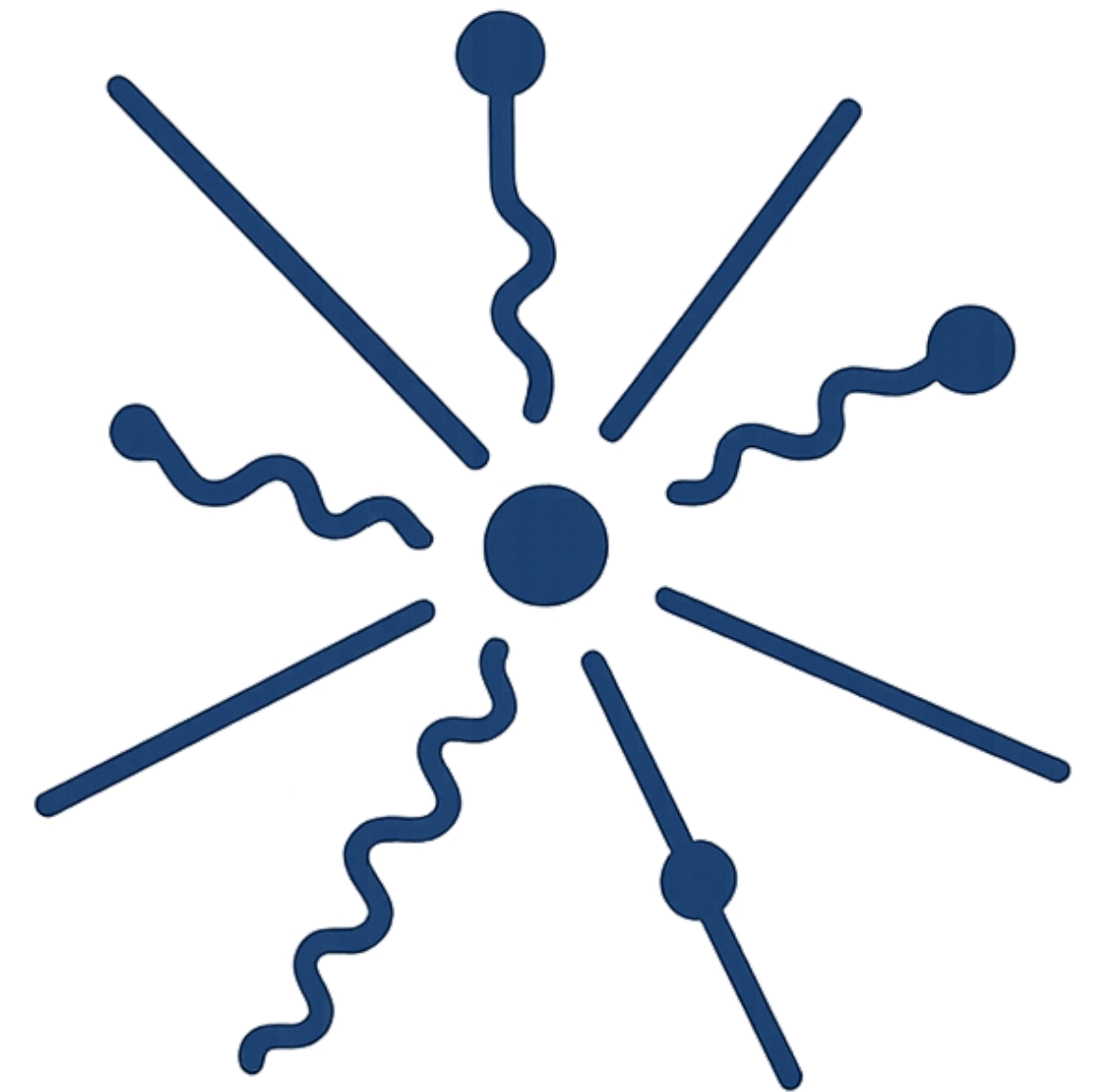
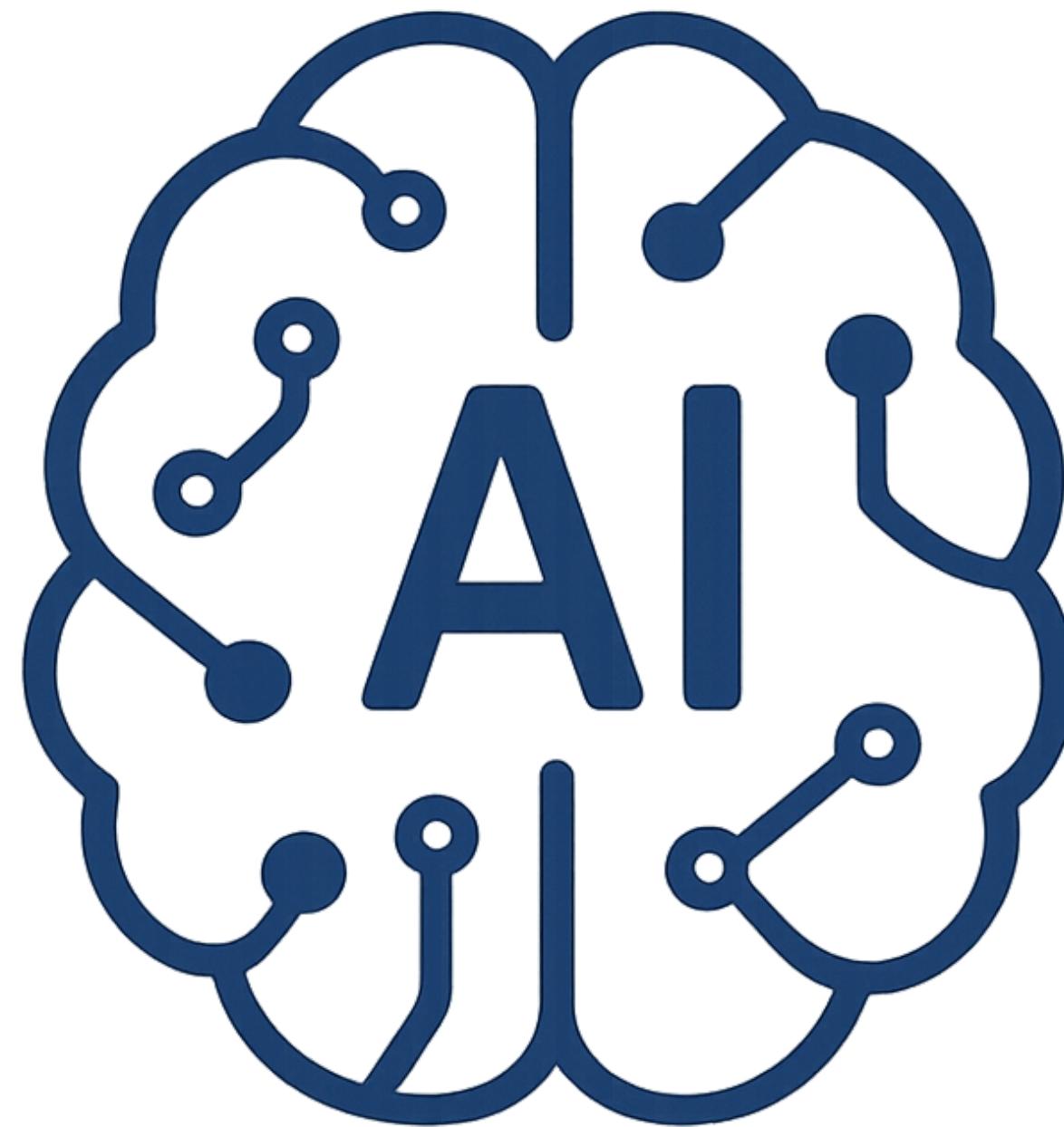
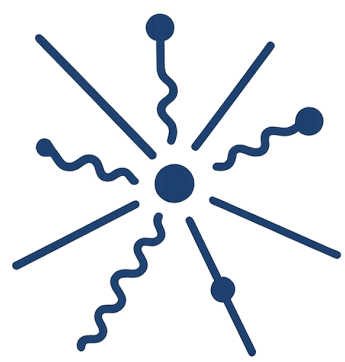
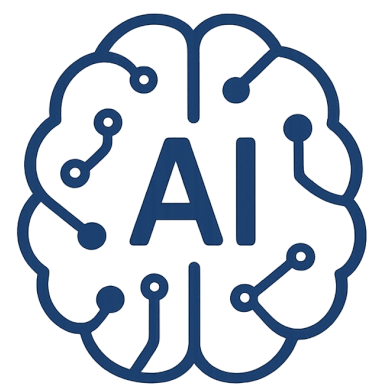


Introduction to

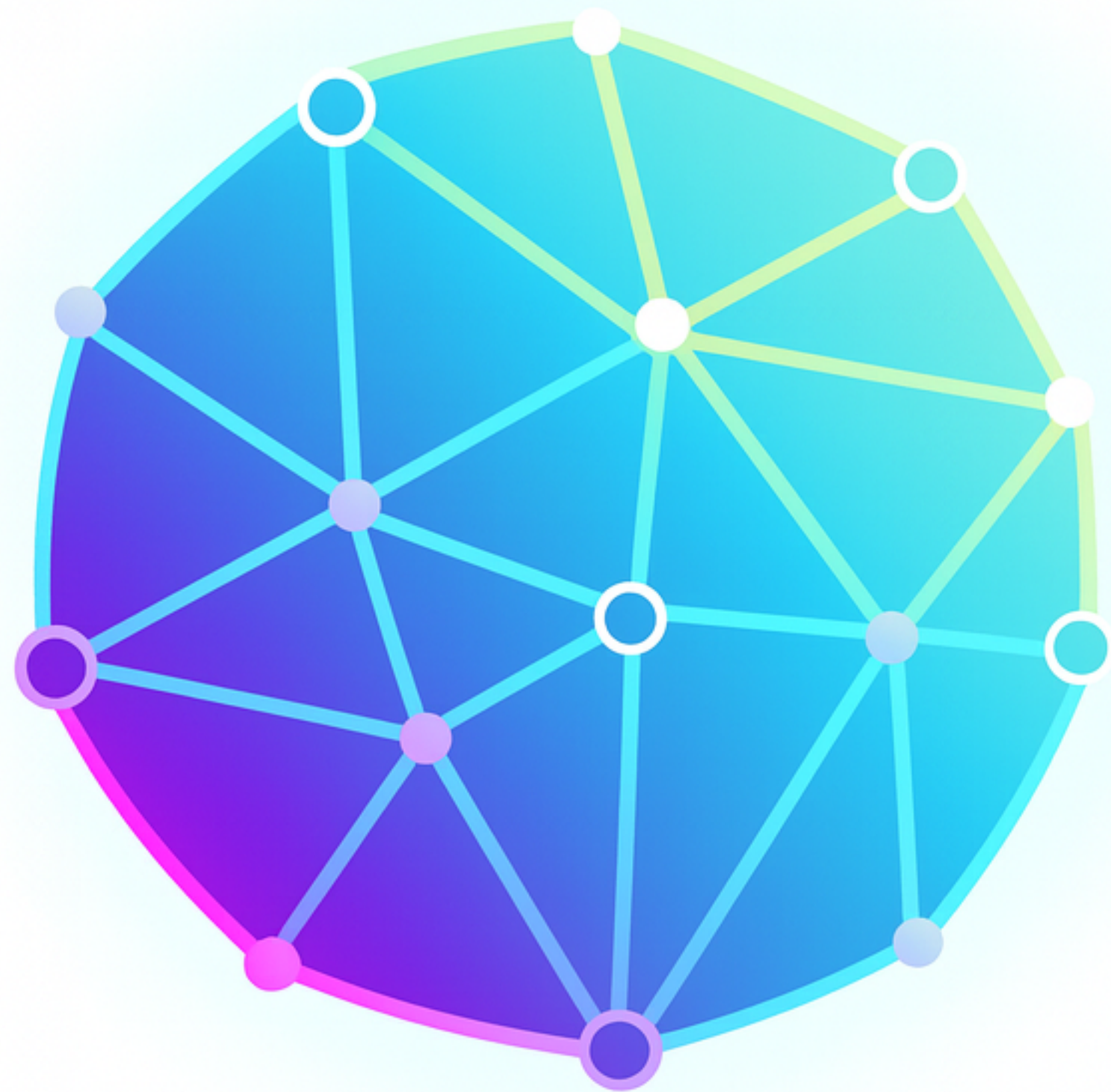
AI-Driven HEP

S. A. Fard - School of Physics (IPM)



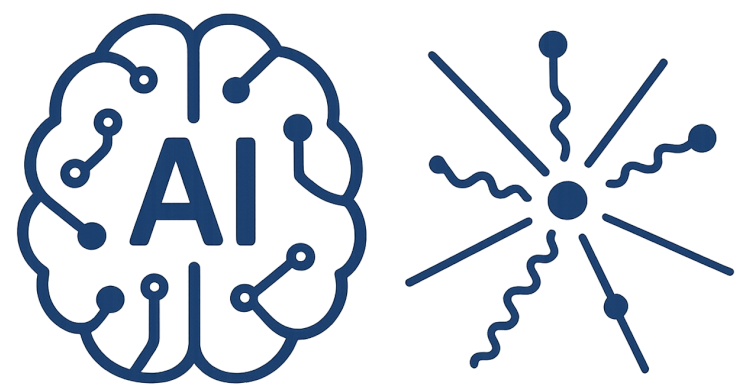


AI-Driven HEP 16



Session 16

Foundation Models



AI-Driven HEP 16

Reference

- [1] Bishop, Christopher M., and Hugh Bishop. Deep learning: Foundations and concepts. Springer Nature, 2023.
- [2] Chollet, Francois. Deep learning with Python. (2021)

<https://physics.ipm.ac.ir/~ansarifard/>

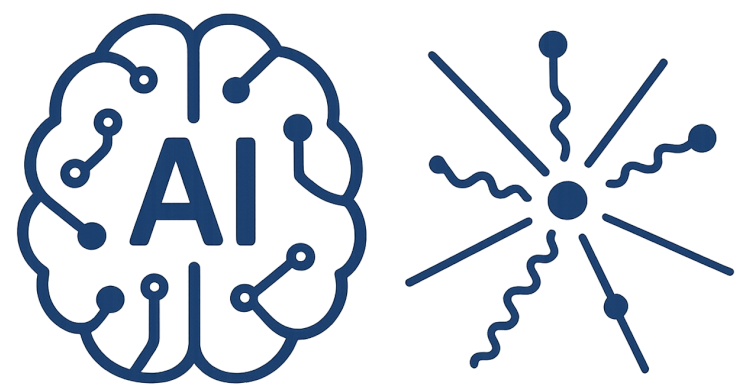


DEEP LEARNING with Python

SECOND EDITION

François Chollet





Reference

- [1] Bishop, Christopher M., and Hugh Bishop. Deep learning: Foundations and concepts. Springer Nature, 2023.
- [2] Chollet, Francois. Deep learning with Python. (2021)

<https://physics.ipm.ac.ir/~ansarifard/>



“Habit of a Persian Lady in 1568.”,
Thomas Jefferys, A
Collection of the
Dresses of Different
Nations, Ancient and
Modern (four
volumes), London,
published between
1757 and 1772

DEEP LEARNING with Python

SECOND EDITION

François Chollet





What we have learned so far

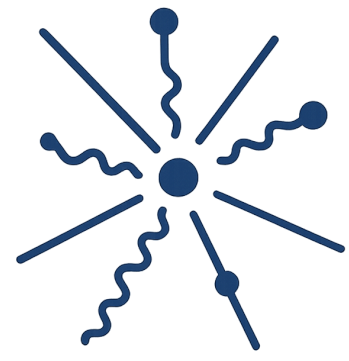
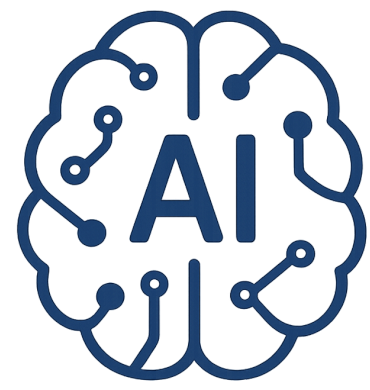
Input data in natural representation



What we have learned so far

Input data in natural representation

Tables, Images, Clean Data,
Languages, Raw Data



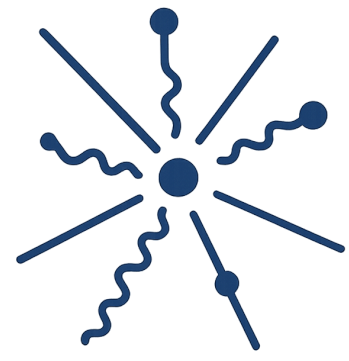
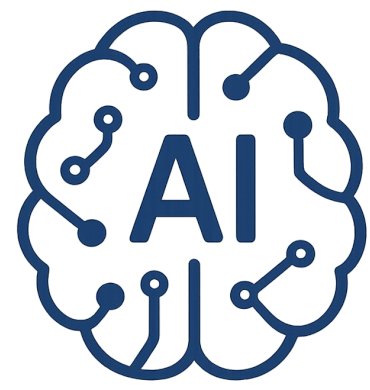
AI-Driven HEP 16

What we have learned so far

Input data in natural representation

As a physicist studying the natural representation of input data is much harder

Tables, Images, Clean Data,
Languages, Raw Data



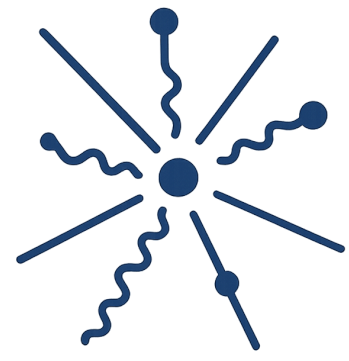
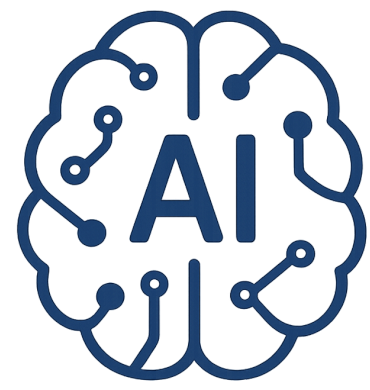
What we have learned so far

Input data in natural representation

As a physicist studying the natural representation of input data is much harder

Tables, Images, Clean Data,
Languages, Raw Data

Traditional view reduces the natural representation to a summary statistics



AI-Driven HEP 16

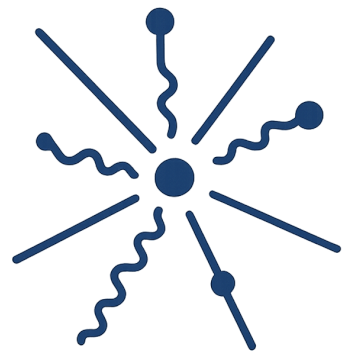
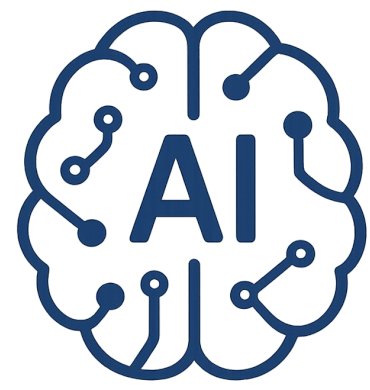
What we have learned so far

Input data in natural representation

Unstructured data (n, k)

Apartment Prices, Galaxy
Categories, Particle Quantum
Numbers

FC-DNN : Transform in Hidden Layers to extract the functional form of the input



AI-Driven HEP 16

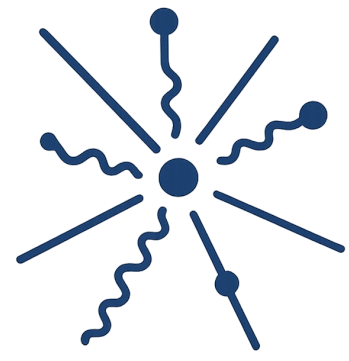
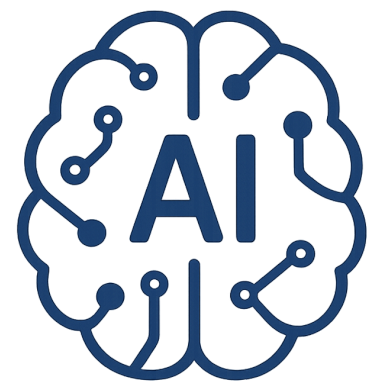
What we have learned so far

Input data in natural representation

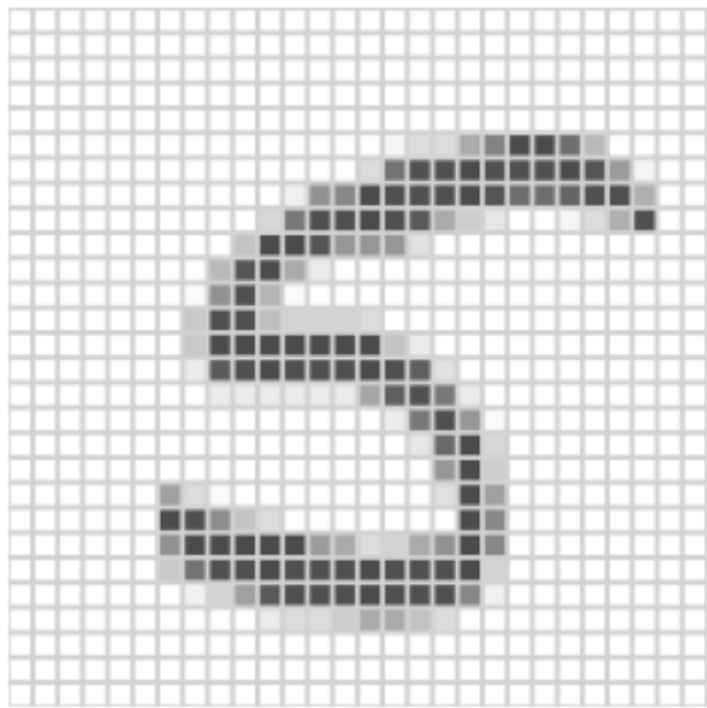
Structured data (n, m, l, k)

Images, Audios, Spectrums,
Particle tracks

CNN : Transform in Hidden Layers to extract local features



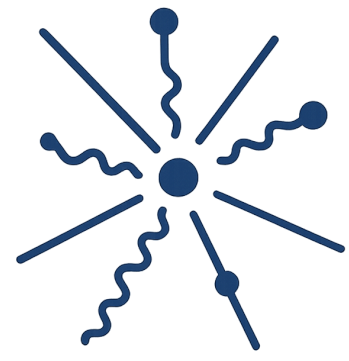
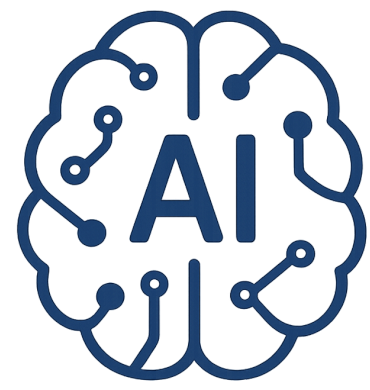
Input as a Sequence



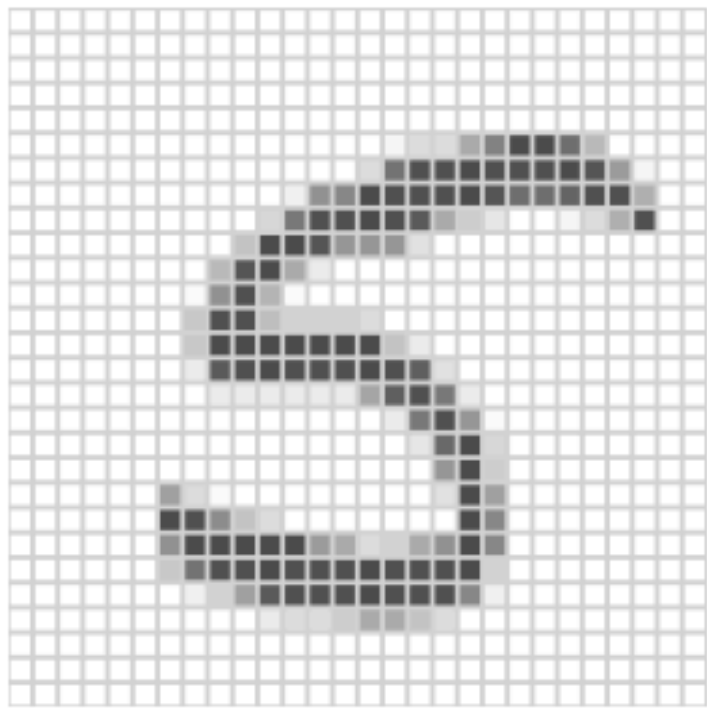
If you remember at first we flatten the images and feed to network

The network have to learn the structure

It has cost when structures are too complex



Input as a Sequence

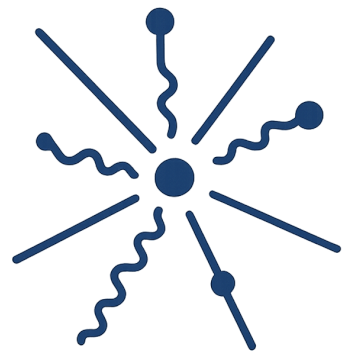
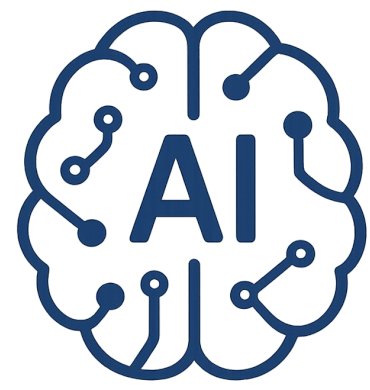


If you remember at first we flatten the images and feed to network

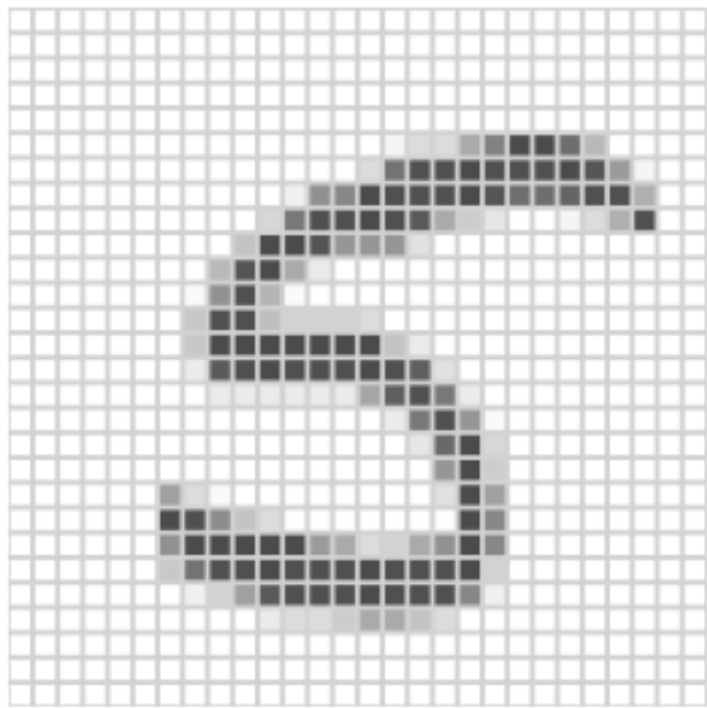
The network have to learn the structure

It has cost when structures are too complex

What if we flatten the structure and feed as a sequence to the network?



Input as a Sequence



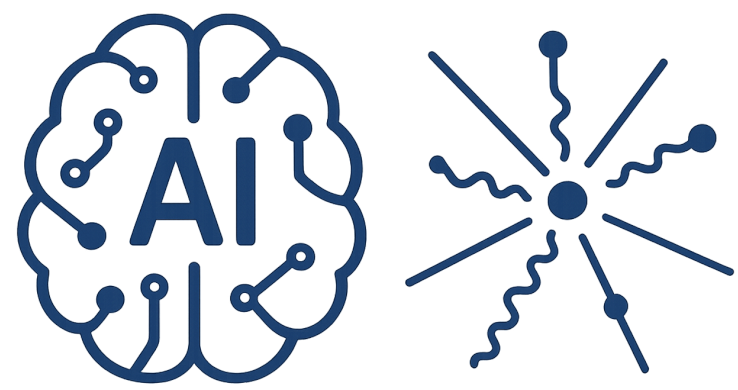
If you remember at first we flatten the images and feed to network

The network have to learn the structure

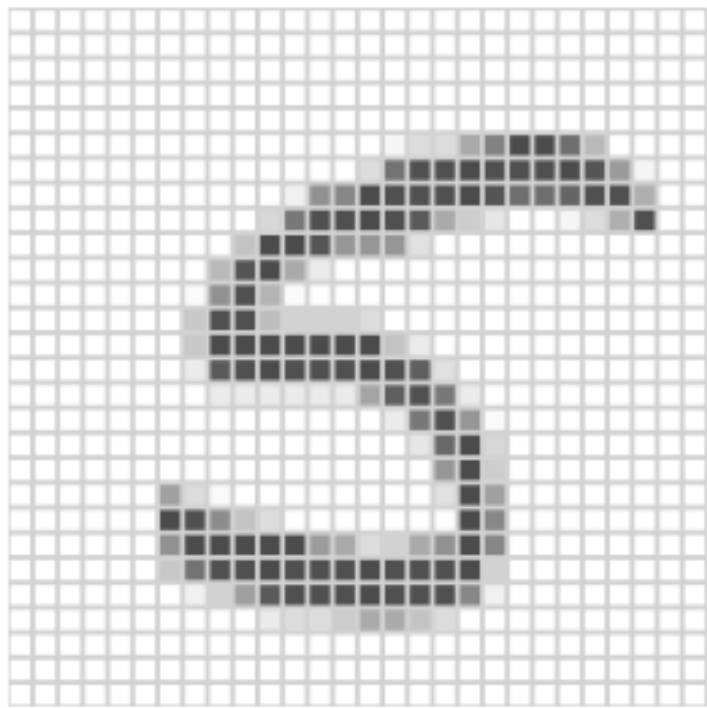
It has cost when structures are too complex

What if we flatten the structure and feed as a sequence to the network?

Suggest simplest model to predict temperature for tomorrow ?



Input as a Sequence



If you remember at first we flatten the images and feed to network

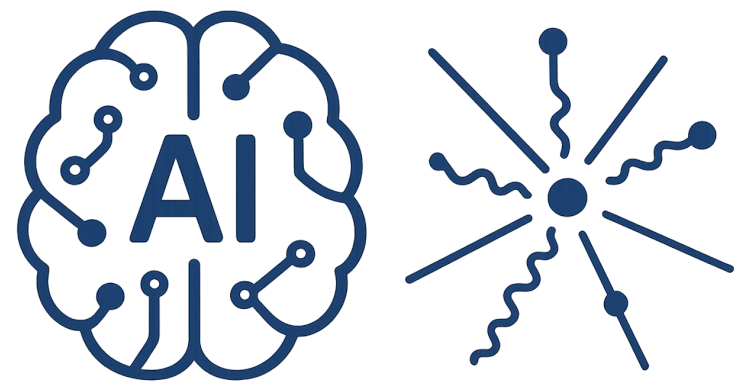
The network have to learn the structure

It has cost when structures are too complex

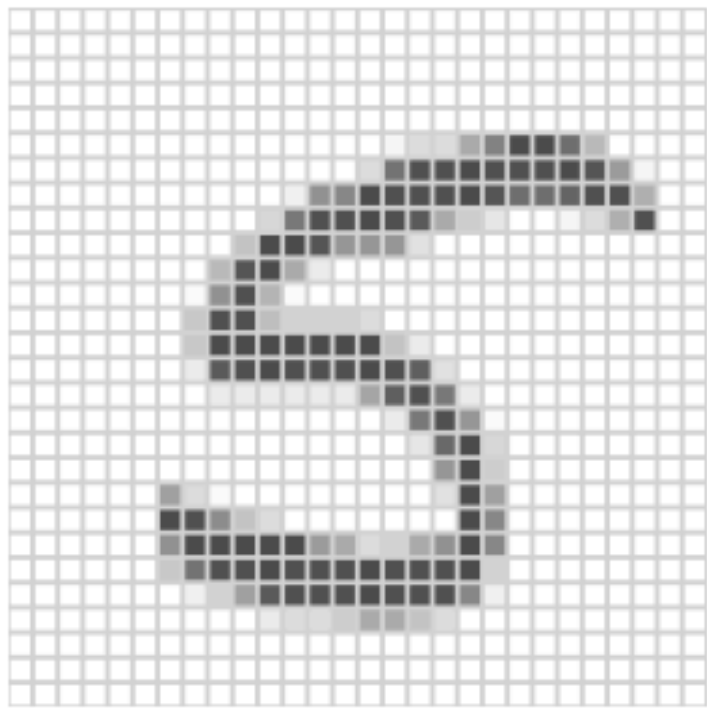
What if we flatten the structure and feed as a sequence to the network?

Suggest simplest model to predict temperature for tomorrow ?

Average of yesterday!



Input as a Sequence



If you remember at first we flatten the images and feed to network

The network have to learn the structure

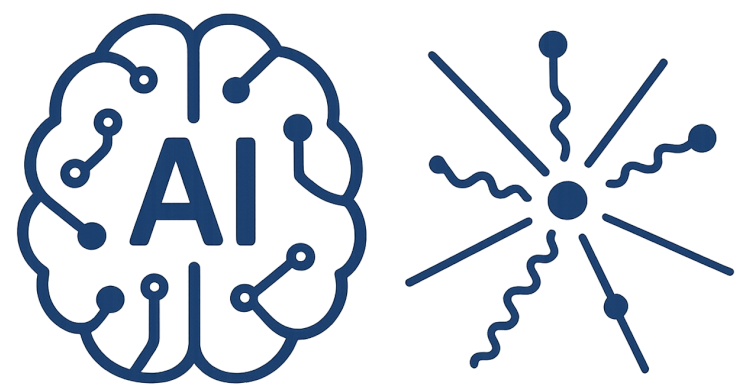
It has cost when structures are too complex

What if we flatten the structure and feed as a sequence to the network?

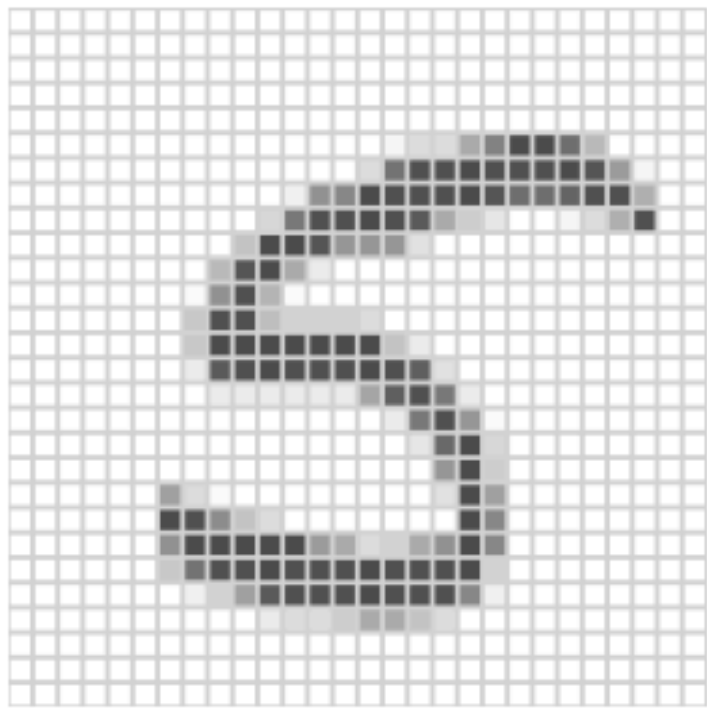
Suggest simplest model to predict temperature for tomorrow ?

Average of yesterday!

Add modification from yearly trend



Recurrent Neural Network

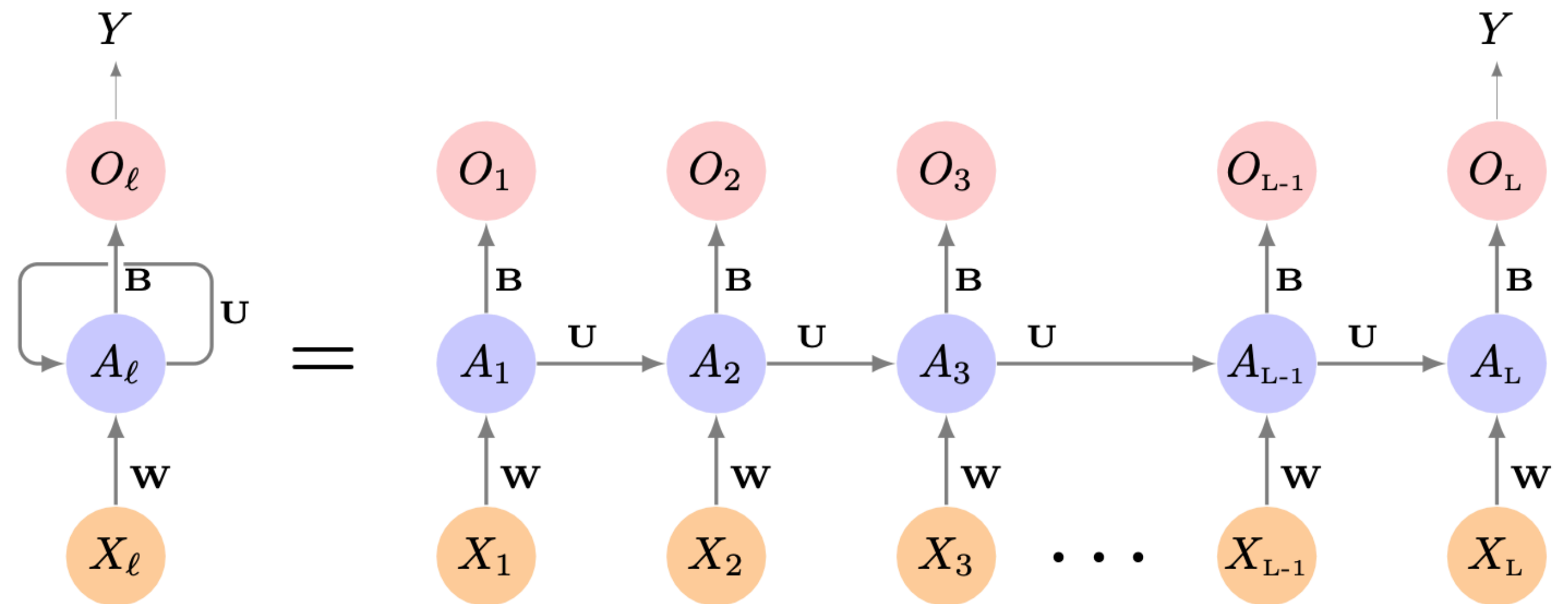


If you remember at first we flatten the images and feed to network

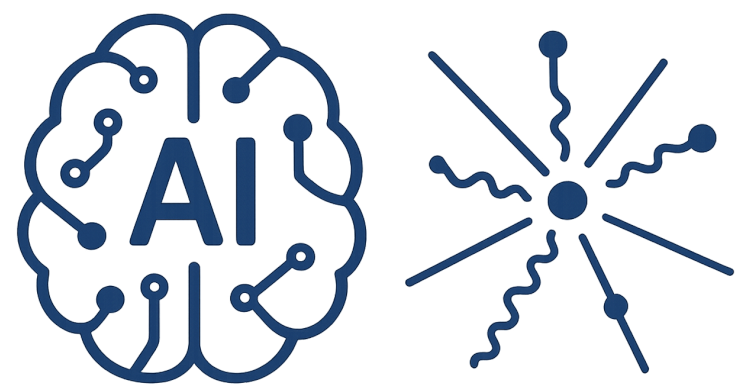
What if we flatten the structure and feed as a sequence to the network?

The network have to learn the structure

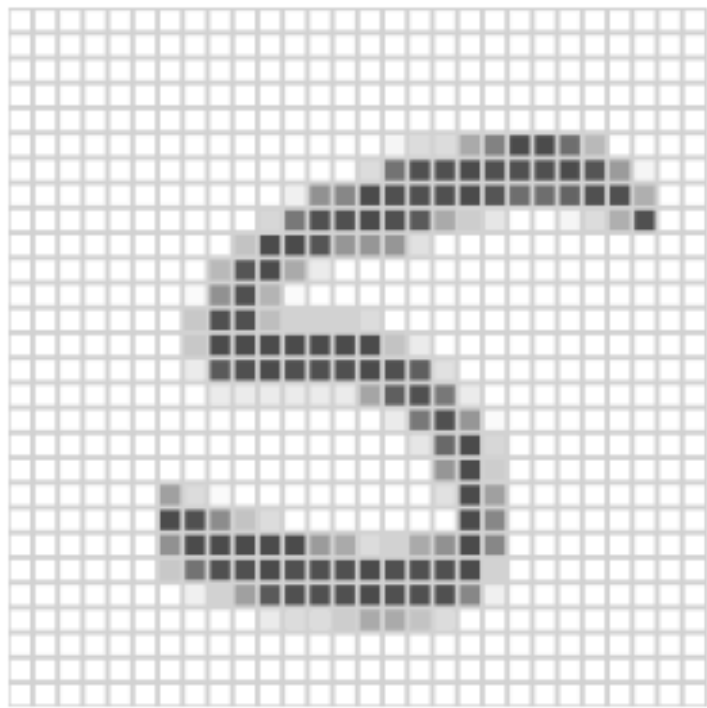
It has cost when structures are too complex



$$A_l = f(Wx_l + UA_{l-1} + B)$$



Recurrent Neural Network



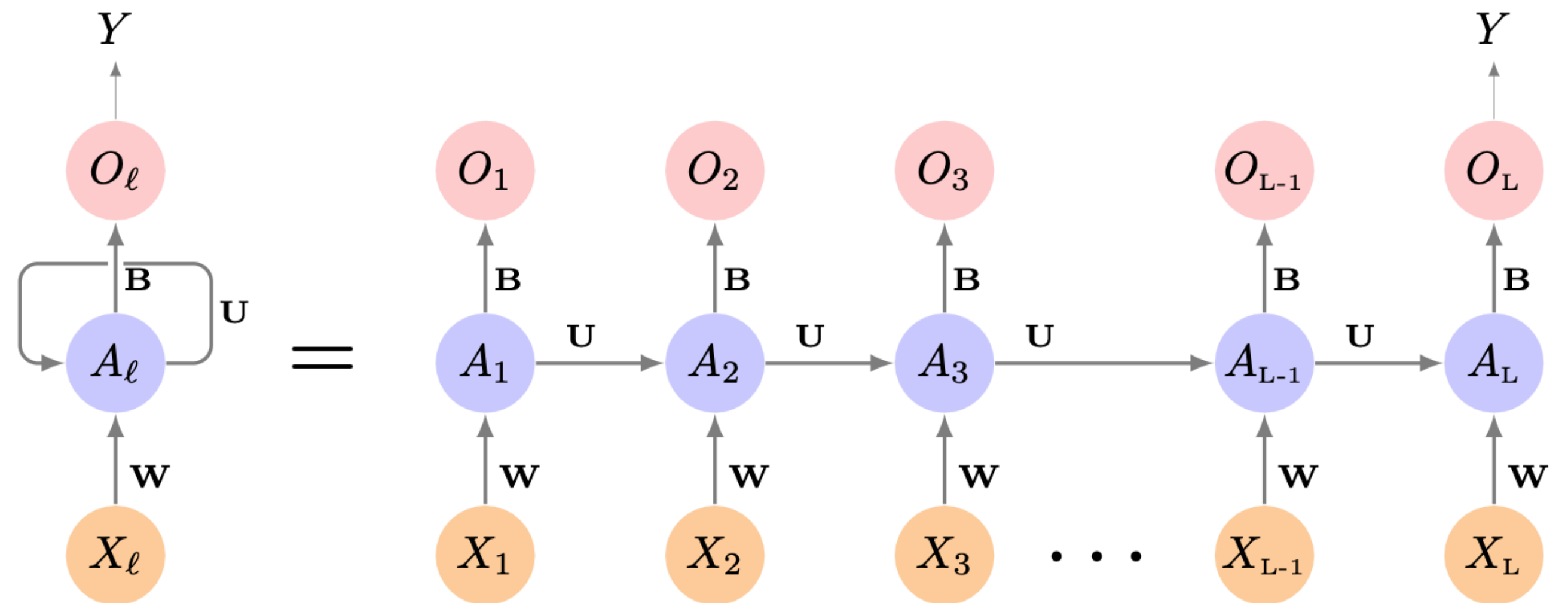
If you remember at first we flatten the images and feed to network

What if we flatten the structure and feed as a sequence to the network?

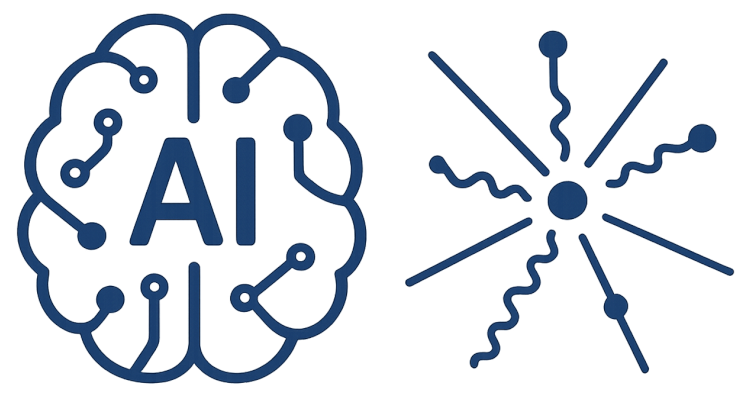
Long term Memory issue

The network have to learn the structure

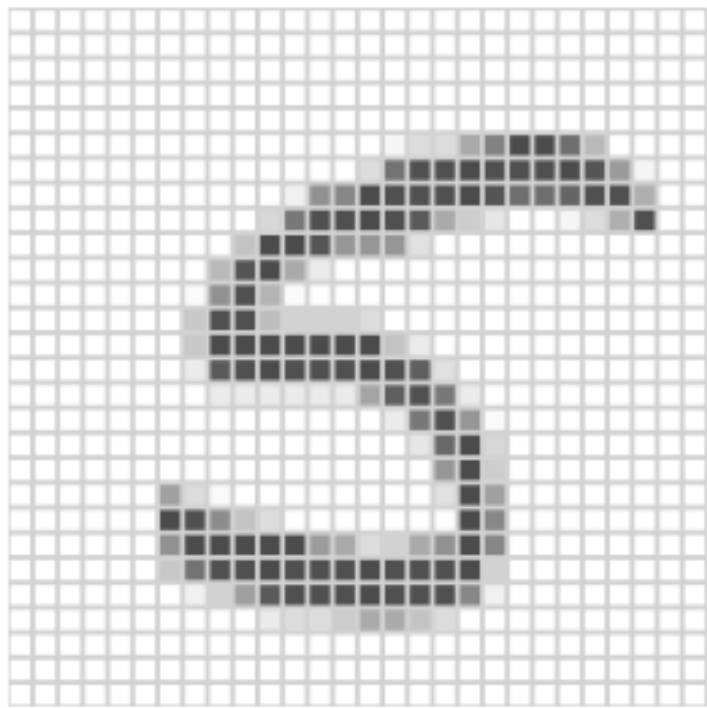
It has cost when structures are too complex



$$A_l = f(Wx_l + UA_{l-1} + B)$$



Recurrent Neural Network



If you remember at first we flatten the images and feed to network

The network have to learn the structure

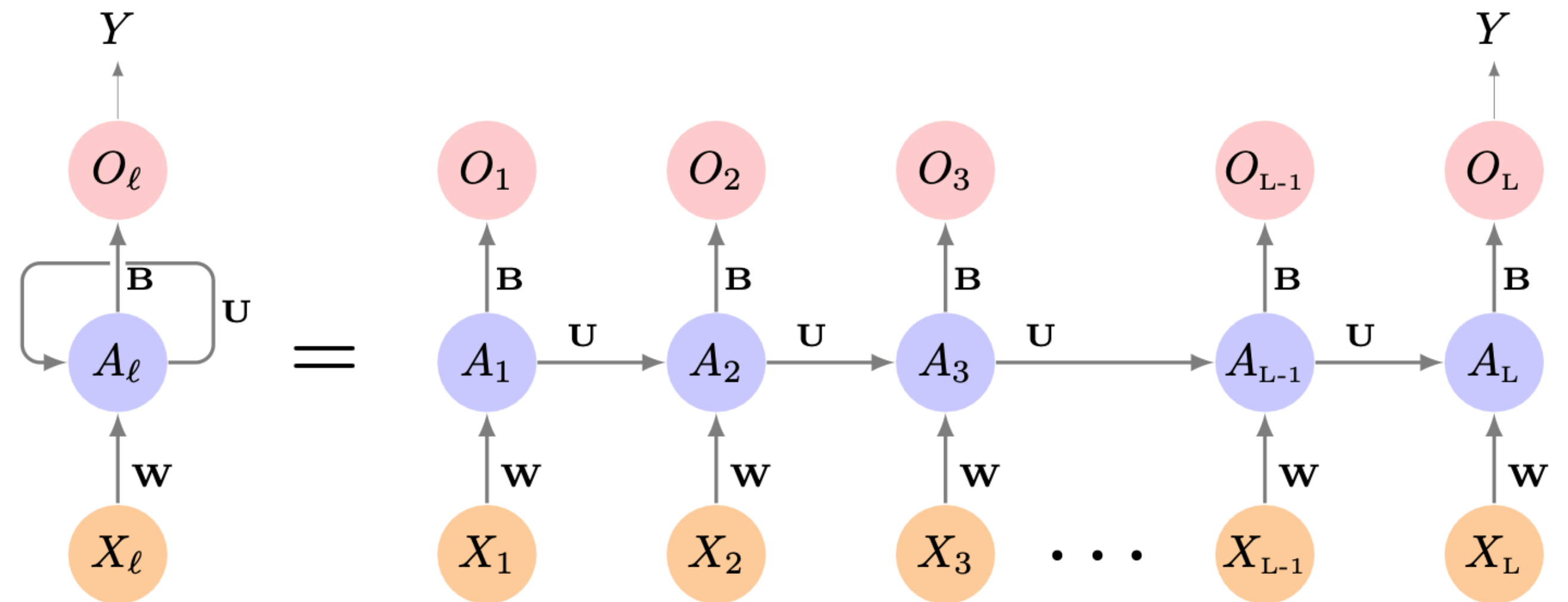
It has cost when structures are too complex

What if we flatten the structure and feed as a sequence to the network?

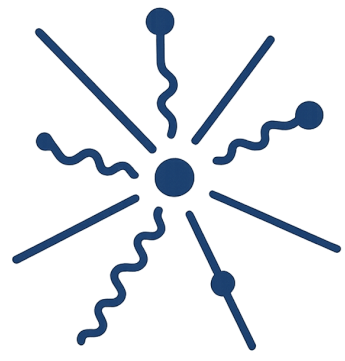
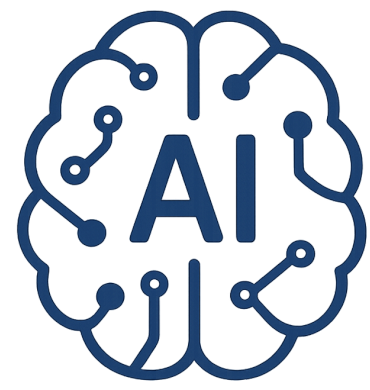
Long term Memory issue

LSTM : An Image with shape (n, s, m, l, k)

Parallel Programming Issue



$$A_l = f(Wx_l + UA_{l-1} + B)$$



Supervised Learning

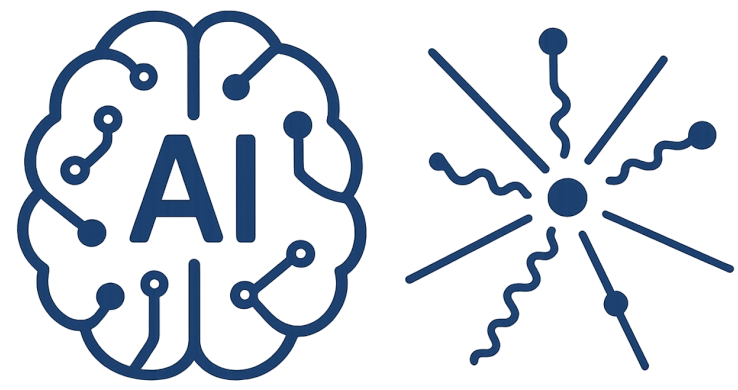
Labels or Targets are known

Hard to Scale

We need architecture that extract global features (context) of structure data

Easy at scale

Parallel running

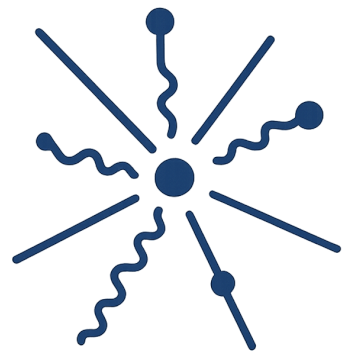
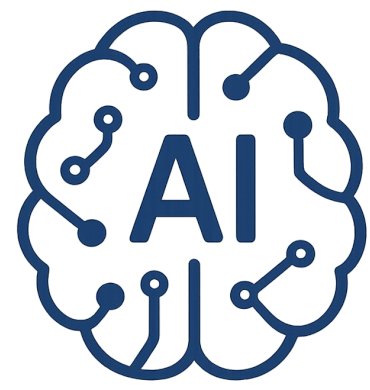


AI-Driven HEP 16

Transformer

One of the most important developments in deep learning

It is Based on a processing concept called Attention

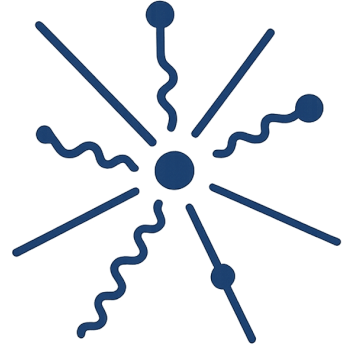
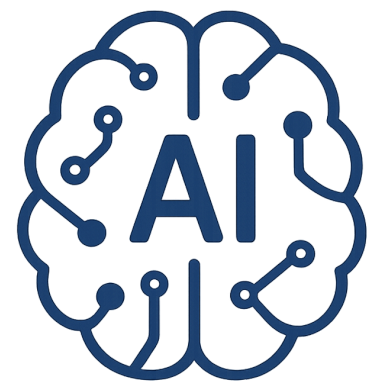


AI-Driven HEP 16

Transformer

Let us play a test

Name the first word bring to your mind when you heard ?



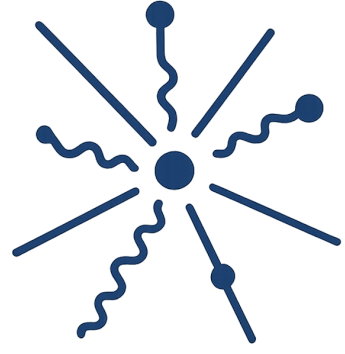
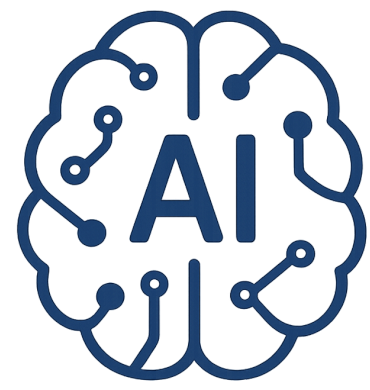
AI-Driven HEP 16

Transformer

Let us play a test

Name the first word bring to your mind when you heard ?

“... The Higgs boson was discovered ...”



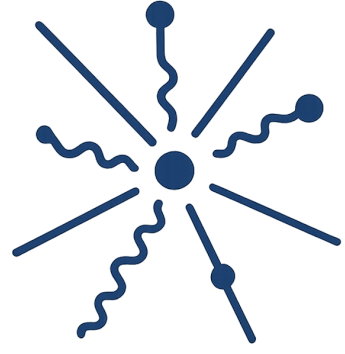
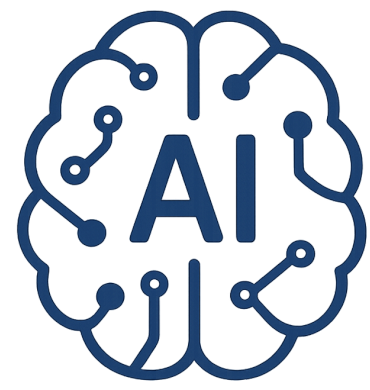
AI-Driven HEP 16

Transformer

Let us play a test

Name the first word bring to your mind when you heard ?

“... The Higgs boson was discovered ...” CERN (LHC)

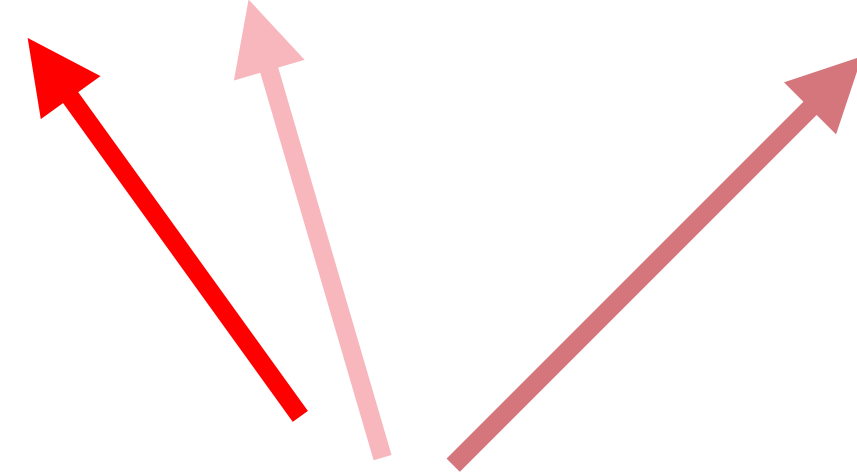


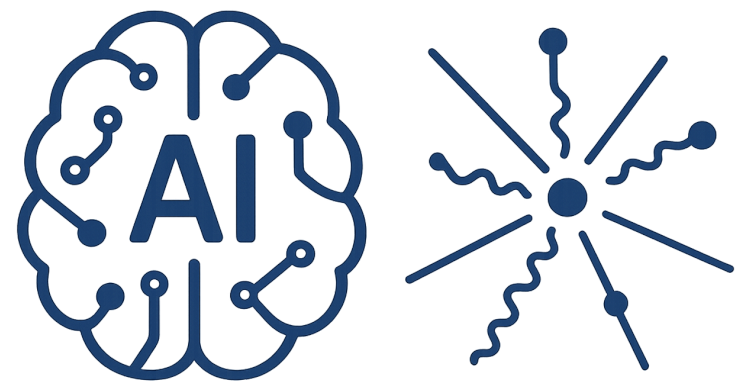
Transformer

Let us play a test

Name the first word bring to your mind when you heard ?

"... The Higgs boson was discovered ..." CERN (LHC)





Transformer

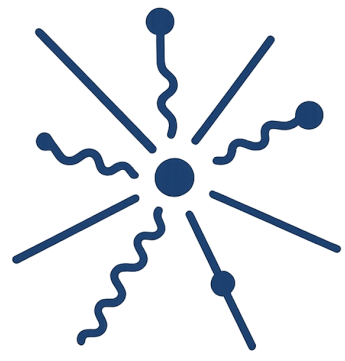
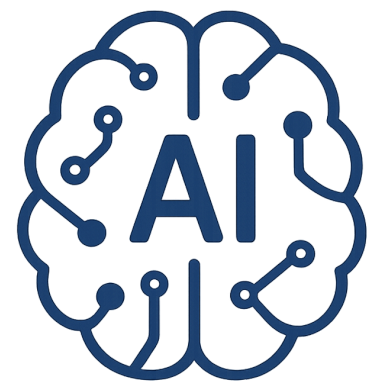
Tokenize building block of your semantic world in D dimensions

discovering → "disc" , "over" , "ing"

discovering dark matter dark → "dark"

matter → "mat", "ter"

Assign a D-vector to each vocabulary

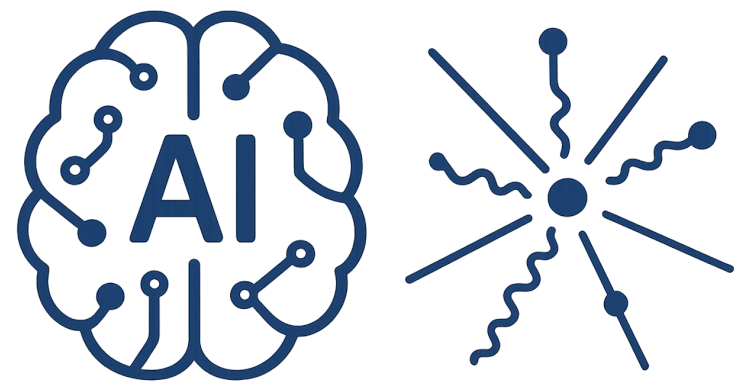


Transformer

Tokenize building block of your semantic world in D dimensions

Make with N numbers of these
building blocks the simplest organs of
your world

$$X : N \times D$$



Transformer

Tokenize building block of your semantic world in D dimensions

Make with N numbers of these building blocks the simplest organs of your world

$$X : N \times D$$

Your training data consists of lots of X

Feed the network with input (B, N_B, D)

